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Subject Name: DBMS

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AIM

To design and implement PL/SQL programs utilizing conditional control statements such as IF–ELSE, IF–ELSIF–ELSE, ELSIF ladder, and CASE constructs in order to control the flow of execution based on logical conditions and to analyze decision-making capabilities in PL/SQL blocks.

S/W Requirement:

- Database Management System: PostgreSQL / Oracle Database Express Edition
- Database Administration Tool: pgAdmin

OBJECTIVES:

- To understand and implement conditional control statements in PL/SQL
- To analyze decision-making using IF–ELSE, ELSIF ladder, and CASE statements
- To enhance logical thinking using PL/SQL blocks

PROBLEM STATEMENT:

Develop and execute PL/SQL programs that demonstrate the use of conditional control statements. The programs should employ IF–ELSE, IF–ELSIF–ELSE, ELSIF ladder, and CASE statements to evaluate given conditions and control the flow of execution accordingly.

1. PROBLEM STATEMENT – IF–ELSE STATEMENT

Write a PL/SQL program to check whether a given number is positive or non-positive using the IF–ELSE conditional control statement and display an appropriate message.

PROGRAM:

DECLARE

num NUMBER := -5;

BEGIN

IF num > 0 THEN

DBMS_OUTPUT.PUT_LINE('The number is Positive');

ELSE

DBMS_OUTPUT.PUT_LINE('The number is Non-Positive');

END IF;

END;

2. PROBLEM STATEMENT – IF–ELSIF–ELSE STATEMENT

Write a PL/SQL program to evaluate the grade of a student based on obtained marks and display the corresponding grade.

PROGRAM:

DECLARE

marks NUMBER := 78;

BEGIN

IF marks >= 90 THEN

DBMS_OUTPUT.PUT_LINE('Grade: A');

ELSIF marks >= 75 THEN

DBMS_OUTPUT.PUT_LINE('Grade: B');

ELSIF marks >= 60 THEN

DBMS_OUTPUT.PUT_LINE('Grade: C');

ELSE

DBMS_OUTPUT.PUT_LINE('Grade: Fail');

END IF;

END;

3. PROBLEM STATEMENT – ELSIF LADDER

Write a PL/SQL program to determine the performance status of a student based on marks using an ELSIF ladder.

PROGRAM:

```
DECLARE
    marks NUMBER := 82;
BEGIN
    IF marks >= 85 THEN
        DBMS_OUTPUT.PUT_LINE('Performance: Excellent');
    ELSIF marks >= 70 THEN
        DBMS_OUTPUT.PUT_LINE('Performance: Very Good');
    ELSIF marks >= 55 THEN
        DBMS_OUTPUT.PUT_LINE('Performance: Good');
    ELSIF marks >= 40 THEN
        DBMS_OUTPUT.PUT_LINE('Performance: Average');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Performance: Poor');
    END IF;
END;
```

4. PROBLEM STATEMENT – CASE STATEMENT

Write a PL/SQL program to display the name of the day based on a given day number using the CASE statement.

PROGRAM:

```
DECLARE
    day_num NUMBER := 3;
    day_name VARCHAR2(20);
BEGIN
    CASE day_num
        WHEN 1 THEN day_name := 'Sunday';
        WHEN 2 THEN day_name := 'Monday';
        WHEN 3 THEN day_name := 'Tuesday';
```

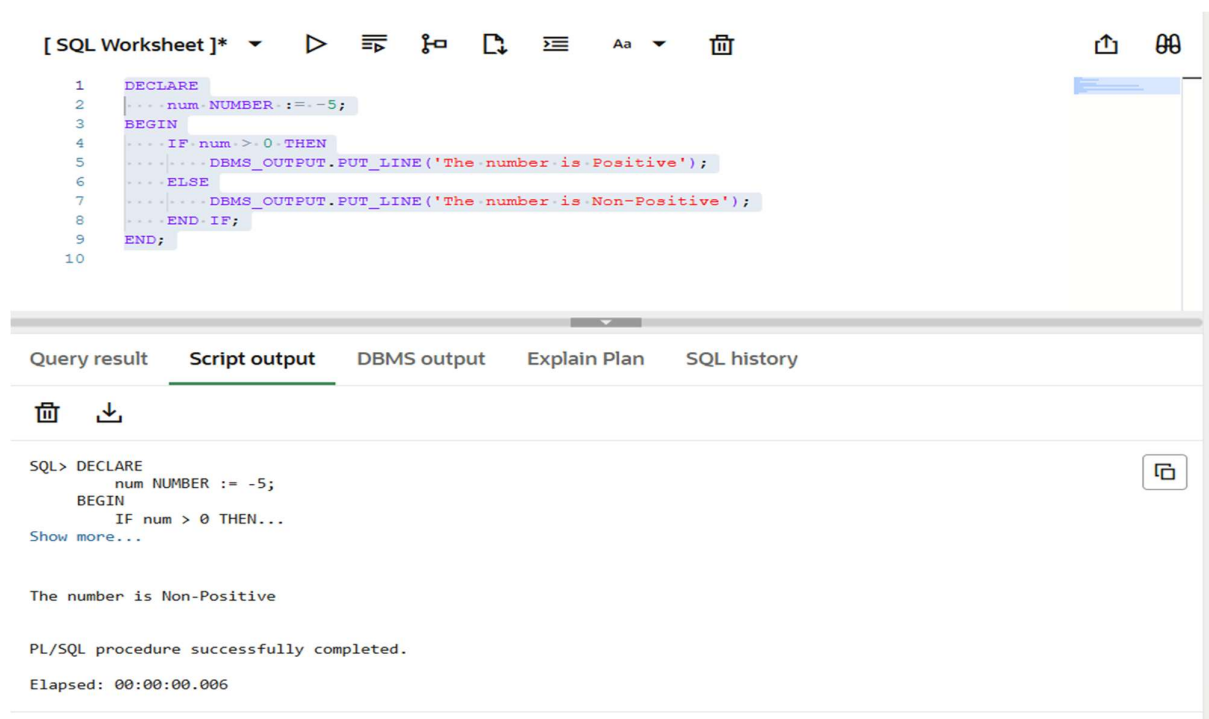
```
WHEN 4 THEN day_name := 'Wednesday';  
WHEN 5 THEN day_name := 'Thursday';  
WHEN 6 THEN day_name := 'Friday';  
WHEN 7 THEN day_name := 'Saturday';  
ELSE day_name := 'Invalid Day Number';  
END CASE;
```

```
DBMS_OUTPUT.PUT_LINE('Day is: ' || day_name);  
END;
```

LEARNING OUTCOMES:

1. Understood the use of conditional control statements in PL/SQL.
2. Learned to apply IF-ELSE and IF-ELSIF-ELSE statements for decision-making.
3. Implemented ELSIF ladder for evaluating multiple conditions.
4. Used CASE statements to simplify complex conditional logic.
5. Improved logical reasoning and procedural programming skills in PL/SQL.

OUTPUT:



The screenshot displays a web-based SQL development environment. The top toolbar includes icons for running queries, saving, and other standard database tools. The main editor area contains the following PL/SQL code:

```
1 DECLARE  
2   num NUMBER := -5;  
3 BEGIN  
4   IF num > 0 THEN  
5     DBMS_OUTPUT.PUT_LINE('The number is Positive');  
6   ELSE  
7     DBMS_OUTPUT.PUT_LINE('The number is Non-Positive');  
8   END IF;  
9 END;  
10
```

Below the editor, a tabbed interface shows the 'Script output' tab selected. The output area displays the execution results:

```
SQL> DECLARE  
      num NUMBER := -5;  
      BEGIN  
      IF num > 0 THEN...  
Show more...  
  
The number is Non-Positive  
  
PL/SQL procedure successfully completed.  
Elapsed: 00:00:00.006
```

[SQL Worksheet]*      Aa 

```

10
11
12 DECLARE
13 ... marks NUMBER := 78;
14 BEGIN
15 ... IF marks >= 90 THEN
16 ... DBMS_OUTPUT.PUT_LINE('Grade: A');
17 ... ELSIF marks >= 75 THEN
18 ... DBMS_OUTPUT.PUT_LINE('Grade: B');
19 ... ELSIF marks >= 60 THEN
20 ... DBMS_OUTPUT.PUT_LINE('Grade: C');
21 ... ELSE
22 ... DBMS_OUTPUT.PUT_LINE('Grade: Fail');
23 ... END IF;
24 END;
25
26

```

Query result **Script output** DBMS output Explain Plan SQL history



Elapsed: 00:00:00.006

```

SQL> DECLARE
      marks NUMBER := 78;
      BEGIN
      IF marks >= 90 THEN...
Show more...

```

Grade: B

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.007

[SQL Worksheet]*      Aa 

```

29
30 DECLARE
31 ... marks NUMBER := 82;
32 BEGIN
33 ... IF marks >= 85 THEN
34 ... DBMS_OUTPUT.PUT_LINE('Performance: Excellent');
35 ... ELSIF marks >= 70 THEN
36 ... DBMS_OUTPUT.PUT_LINE('Performance: Very Good');
37 ... ELSIF marks >= 55 THEN
38 ... DBMS_OUTPUT.PUT_LINE('Performance: Good');
39 ... ELSIF marks >= 40 THEN
40 ... DBMS_OUTPUT.PUT_LINE('Performance: Average');
41 ... ELSE
42 ... DBMS_OUTPUT.PUT_LINE('Performance: Poor');
43 ... END IF;
44 END;
45

```

Query result **Script output** DBMS output Explain Plan SQL history



Elapsed: 00:00:00.007

```

SQL> DECLARE
      marks NUMBER := 82;
      BEGIN
      IF marks >= 85 THEN...
Show more...

```

Performance: Very Good

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008

[SQL Worksheet]*      Aa   

```
47 /
48 DECLARE
49     day_num NUMBER := 3;
50     day_name VARCHAR2(20);
51 BEGIN
52     CASE day_num
53     WHEN 1 THEN day_name := 'Sunday';
54     WHEN 2 THEN day_name := 'Monday';
55     WHEN 3 THEN day_name := 'Tuesday';
56     WHEN 4 THEN day_name := 'Wednesday';
57     WHEN 5 THEN day_name := 'Thursday';
58     WHEN 6 THEN day_name := 'Friday';
59     WHEN 7 THEN day_name := 'Saturday';
60     ELSE day_name := 'Invalid Day Number';
61     END CASE;
62
63     DBMS_OUTPUT.PUT_LINE('Day is: ' || day_name);
64 END;
65
66
67
```

Query result **Script output** DBMS output Explain Plan SQL history

```
SQL script
    day_num NUMBER := 3;
    day_name VARCHAR2(20);
    BEGIN...
Show more...
```

Day is: Tuesday

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.012

CONCLUSION:

This experiment provided hands-on experience with conditional control statements in PL/SQL. The use of IF-ELSE, ELSIF ladder, and CASE statements helped in understanding decision-making mechanisms and control flow within PL/SQL programs.